ggplot2 Deep Dive: Aesthetics & Geometries

Workshop 2 — Module 3

Prof.Asoc. Endri Raco

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Data Visualization for Data Scientists
UNYT Tirana

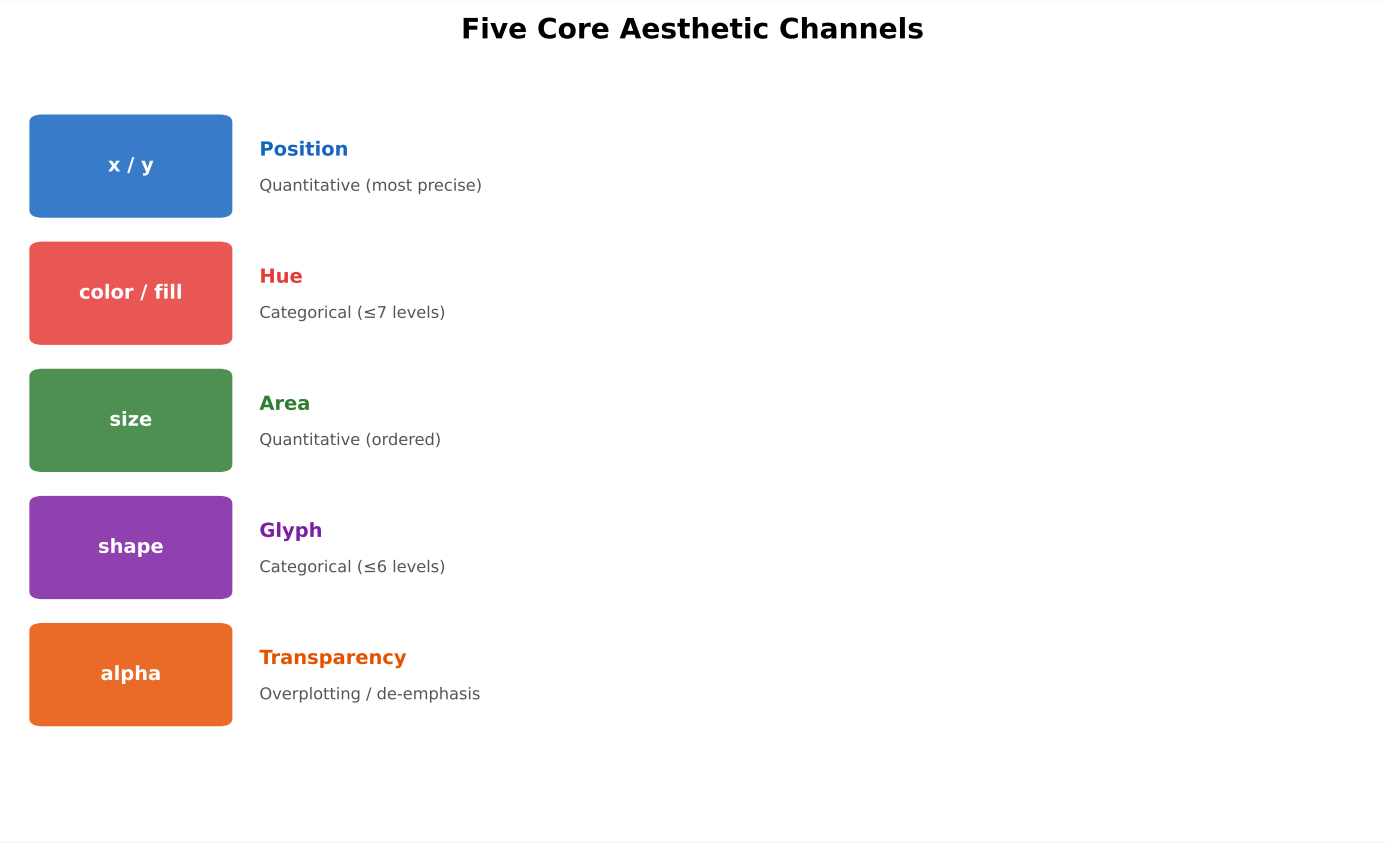
Module Roadmap

1. The five core aesthetic channels: x/y, color, size, shape, alpha
2. Continuous vs discrete colour mapping
3. Layering multiple geometries with +
4. Overplotting: diagnosis and three fixes
5. Special geometries: text, label, segment, lollipop, dumbbell

Learning Outcome
You will be able to encode up to five variables simultaneously using aesthetic channels, layer multiple geoms on one axes, and solve overplotting with alpha, jitter, or hexbin.

The Five Core Aesthetics

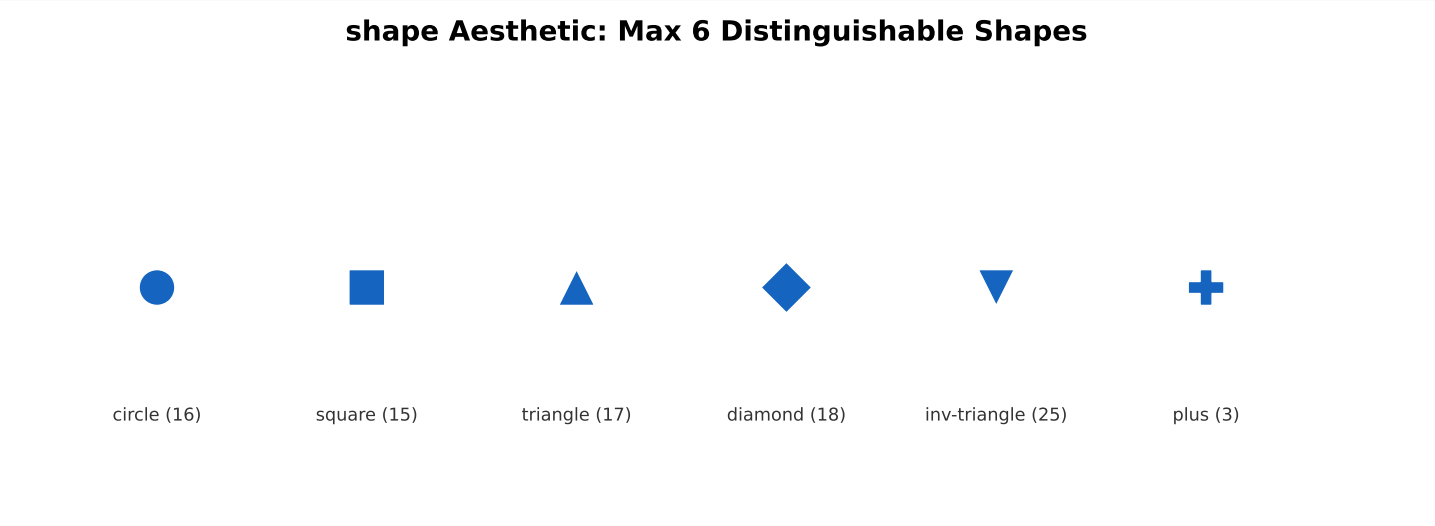
Aesthetic Channels



Pro-Tip

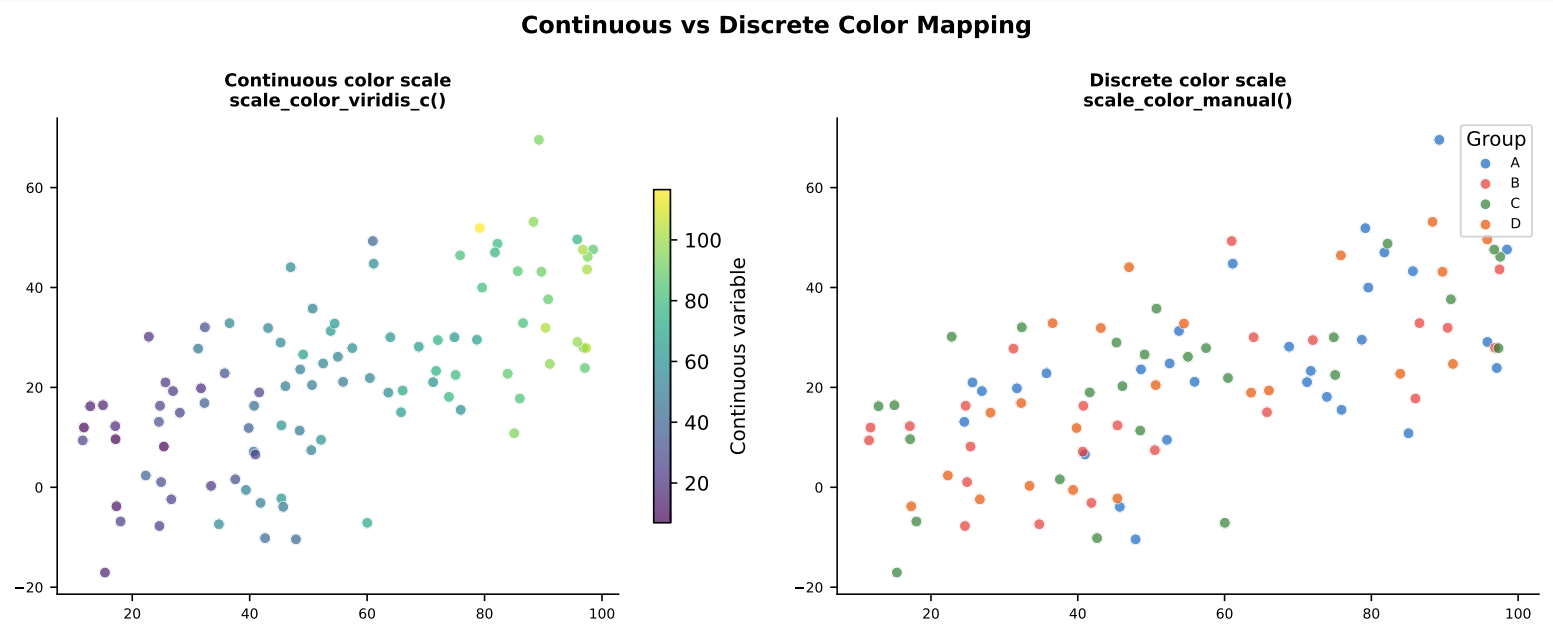
Position (x, y) is the most precise channel (Cleveland rank 1). Colour hue is next for categorical data. Size works for ordered quantitative data but not for precise comparisons. Shape tops out at 6 distinguishable glyphs. Alpha is a last resort.

The shape Palette: Maximum 6 Levels



ggplot2's default shape scale uses filled shapes for the first 6 levels, then switches to unfilled — which become nearly indistinguishable. If your categorical variable has > 6 levels, use colour or faceting instead.

Continuous vs Discrete Colour



Data Insight

Continuous variables map to a colour gradient (`scale_color_viridis_c`). Discrete variables map to a qualitative palette (`scale_color_brewer`). Mismatching produces misleading perceptual ordering.

Multi-Aesthetic Encoding in R

```r
# Five variables in one chart
ggplot(df, aes(
    x     = gdp,          # position
    y     = life_exp,      # position
    color = continent,     # hue
    size  = population,    # area
    shape = income_group # glyph
  )) +
  geom_point(alpha = 0.6) +
  scale_color_brewer(
    palette = "Set2") +
  scale_size(range = c(1, 12)) +
  scale_shape_manual(
    values = c(16, 17, 15)) +
  theme_minimal() +
  labs(title = "Gapminder: 5 Vars")
```

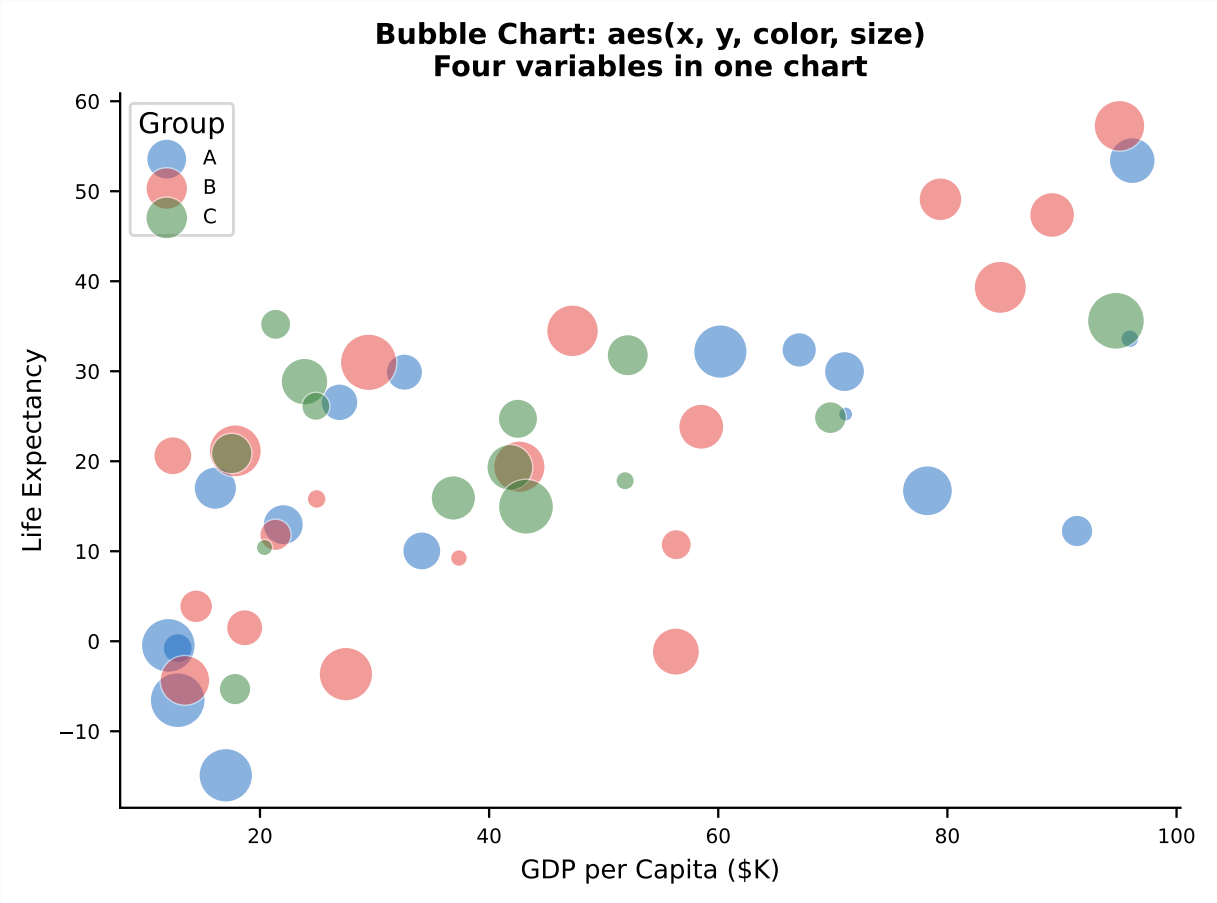
Five variables encoded: x = GDP, y = life expectancy, colour = continent, size = pop, shape = income group. Each `aes()` creates its own legend.

Multi-Aesthetic Encoding in Python

```python
# plotnine: identical syntax
(ggplot(df, aes(
    x="gdp", y="life_exp",
    color="continent",
    size="population",
    shape="income_group"))
 + geom_point(alpha=0.6)
 + scale_color_brewer(
     type="qual", palette="Set2")
 + scale_size(range=(1, 12))
 + theme_minimal())

# matplotlib OO alternative:
for grp in groups:
    mask = df["continent"] == grp
    ax.scatter(
        df.loc[mask, "gdp"],
        df.loc[mask, "life_exp"],
        s=df.loc[mask, "pop"] / 1e5
            ,
        c=palette[grp],
        marker=markers[grp],
        alpha=0.6, label=grp)
```

plotnine handles multi- aesthetic legends automatically. In matplotlib, you must loop over groups and manage legends manually.

Bubble Chart: The size Aesthetic

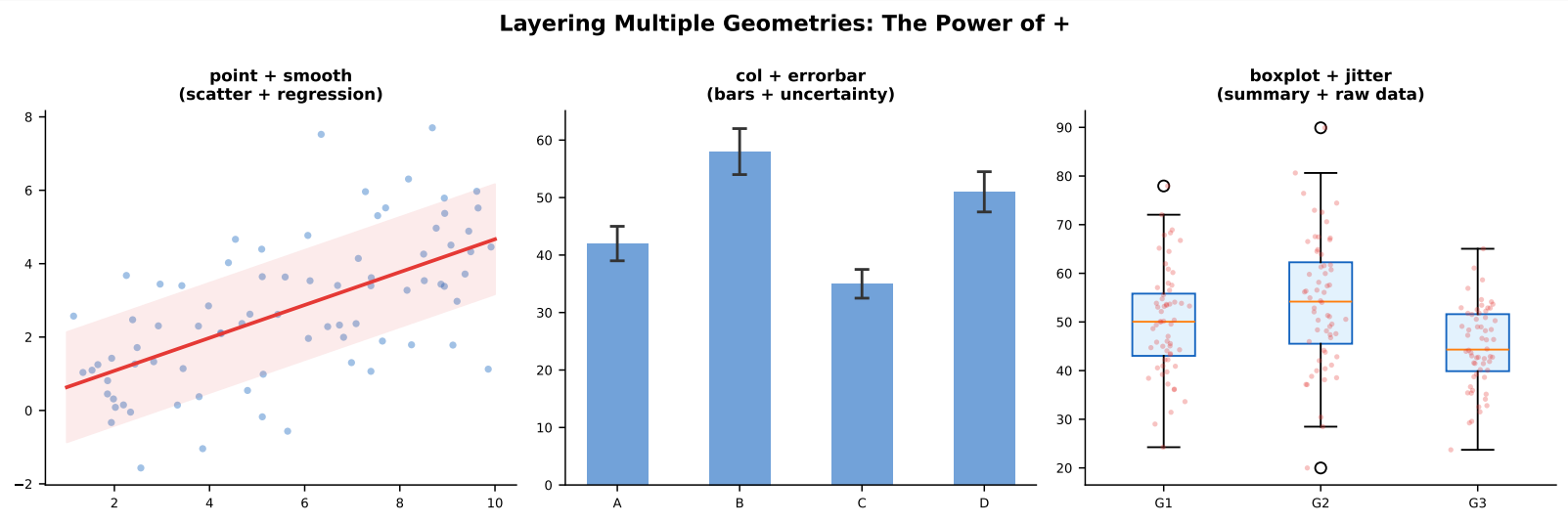


Pro-Tip

`scale_size(range = c(1, 12))` controls the min/max point size. ggplot2 maps to area (perceptually correct). In matplotlib, `s` is in points^2 — already area-based.

Layering Multiple Geometries

The Power of +: Three Geom Combinations



Geom Layering in Code

R: point + smooth

```
ggplot(df, aes(x, y)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm",
    color = "#E53935")
# Two layers: raw data + trend
```

R: boxplot + jitter

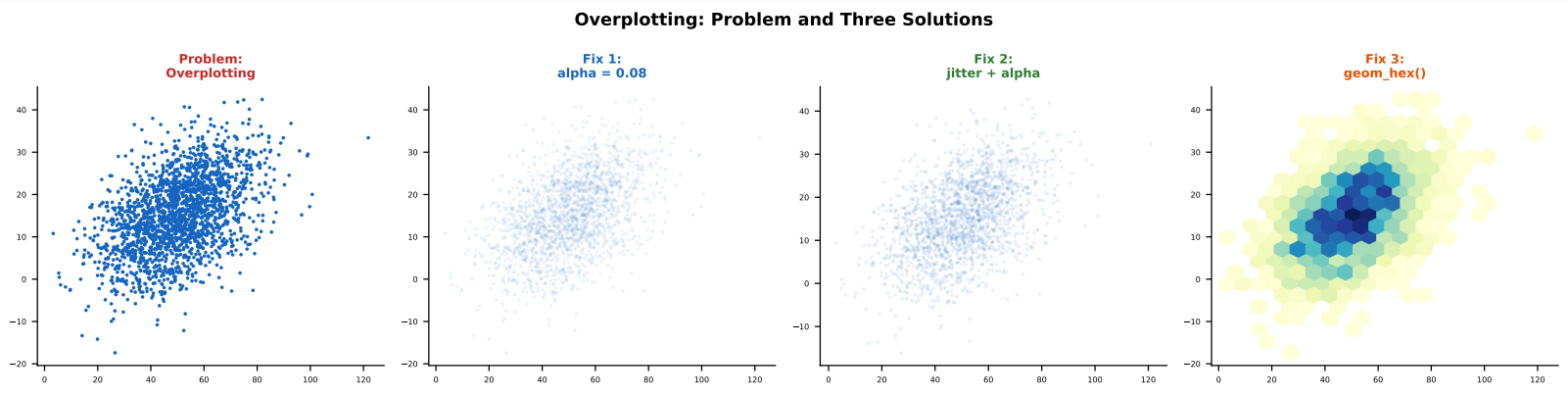
```
ggplot(df, aes(x = group,
               y = value)) +
  geom_boxplot(outlier.shape = NA
      ,
    fill = "#E3F2FD") +
  geom_jitter(width = 0.15,
    alpha = 0.3, size = 1,
    color = "#E53935")
# Show summary AND raw data
```

R: col + errorbar

```
df_summary <- df |>
  group_by(cat) |>
  summarise(
    m = mean(val),
    se = sd(val)/sqrt(n()))

ggplot(df_summary,
  aes(x = cat, y = m)) +
  geom_col(fill = "#1565C0",
    alpha = 0.6, width = 0.5) +
  geom_errorbar(
    aes(ymin = m - se,
        ymax = m + se),
    width = 0.2,
    linewidth = 0.8) +
  coord_flip()
```

Overplotting

Problem and Three Solutions



Overplotting Solutions in R

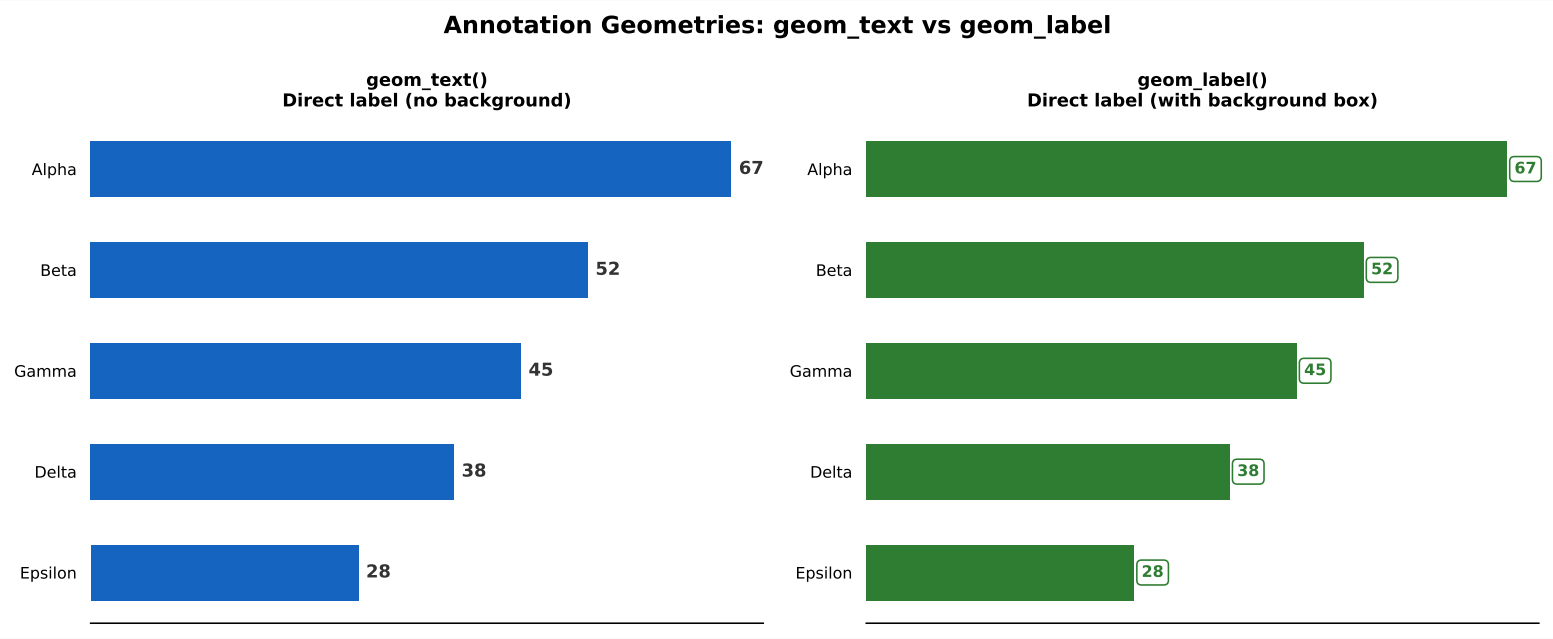
```r
# Fix 1: Reduce alpha (most common)
ggplot(df, aes(x, y)) +
  geom_point(alpha = 0.05, size = 0.5)      # near-transparent

# Fix 2: Jitter (for discrete x)
ggplot(df, aes(x = category, y = value)) +
  geom_jitter(width = 0.2, height = 0, alpha = 0.15, size = 0.8)

# Fix 3: Hexbin (for large n)
ggplot(df, aes(x, y)) +
  geom_hex(bins = 25) +
  scale_fill_viridis_c()                    # count encoded as colour

# Fix 4: 2D density contours
ggplot(df, aes(x, y)) +
  geom_density_2d_filled(alpha = 0.7) +
  geom_point(alpha = 0.02, size = 0.3)      # ghost points for
      context
```

Pro-Tip

Rule of thumb: $n < 500$ use alpha; $500 < n < 5000$ use hex/density; $n > 5000$ use aggregation (geom_bin2d or geom_hex).

Special Geometries

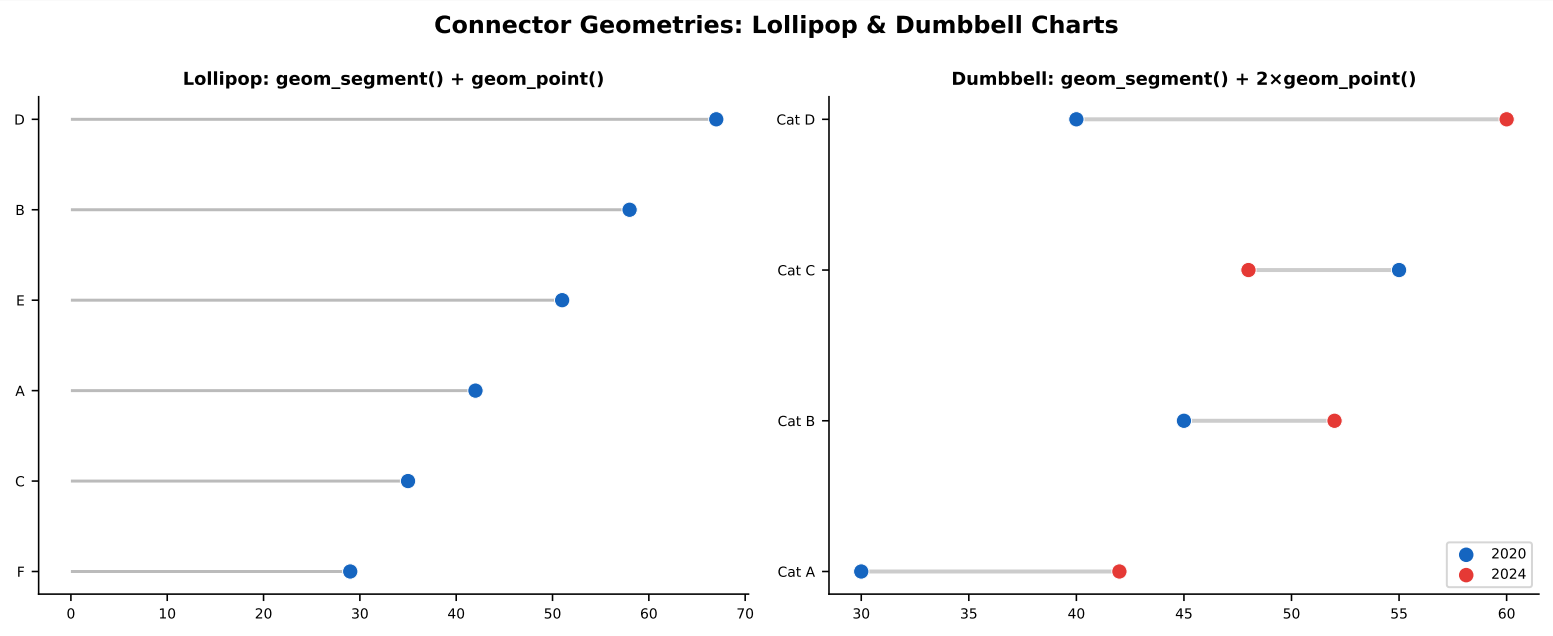
Annotation Geoms: text vs label



Data Insight

`geom_text()` renders plain text (no background). `geom_label()` adds a filled box — better visibility on cluttered backgrounds. For anti-overlap, use `ggrepel::geom_text_repel()` (R) or `adjustText` (Python).

Connector Geoms: Lollipop & Dumbbell



Lollipop & Dumbbell in R

```r
# Lollipop: geom_segment + geom_point
ggplot(df, aes(x = fct_reorder(cat, val), y = val)) +
  geom_segment(aes(xend = cat, y = 0, yend = val), color = "grey70")
      +
  geom_point(size = 3, color = "#1565C0") +
  coord_flip() + theme_minimal()

# Dumbbell: two geom_point layers + geom_segment
ggplot(df, aes(y = fct_reorder(cat, val_2024))) +
  geom_segment(aes(x = val_2020, xend = val_2024,
                   yend = cat), color = "grey80", linewidth = 1.5) +
  geom_point(aes(x = val_2020), size = 3, color = "#1565C0") +
  geom_point(aes(x = val_2024), size = 3, color = "#E53935") +
  theme_minimal() +
  labs(x = "Value", y = NULL, title = "Change: 2020 vs 2024")
```

Synthesis

Aesthetic Selection Checklist

Variable Type	Best Aesthetic	Fallback
Quantitative (primary)	x or y position	size
Quantitative (secondary)	colour gradient	size
Categorical (≤ 6)	colour hue or shape	facet
Categorical (> 6)	facet_wrap	colour + label
Ordered categorical	colour luminance	size
Binary	colour (2 hues)	shape or alpha

Pro-Tip
Encode the most important variable on the highest-precision channel (position). Reserve colour for the second-most-important grouping.

Key Takeaways

1. Five core aesthetics: **x/y** (position), **color** (hue/gradient), **size** (area), **shape** (glyph, ≤ 6), **alpha** (transparency).
2. Match the aesthetic to the variable type: continuous $\rightarrow$ position or gradient; categorical $\rightarrow$ hue or shape.
3. **Layering geoms** with + creates composite charts: point+smooth, boxplot+jitter, col+errorbar.
4. **Overplotting** is solved by alpha ($n < 500$), jitter (discrete x), hex/density ($n > 500$).
5. **Special geoms**: text/label for annotation, segment+point for lollipop and dumbbell charts.

Essential Readings

- Wickham, H. (2016). *ggplot2*, Chapters 4–5 (Mastering Aesthetics, Geom Catalogue).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapters 2, 18 (Aesthetics, Handling Overlapping Points).
- Chang, W. (2024). *R Graphics Cookbook*, 2nd ed., Chapters 5–6 (Scatter Plots, Bar Charts).

Next Module: Scales & Coordinate Systems (W02-M04)